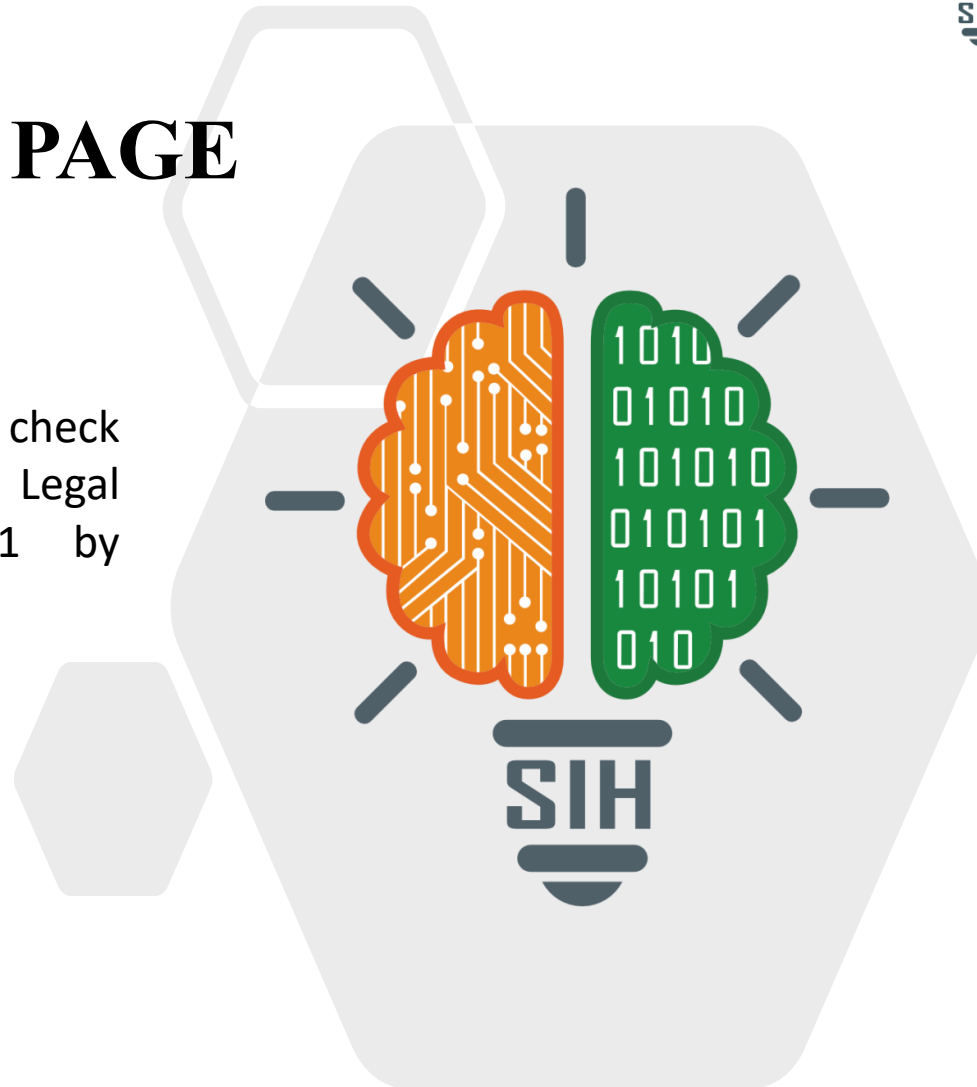


SMART INDIA HACKATHON 2026



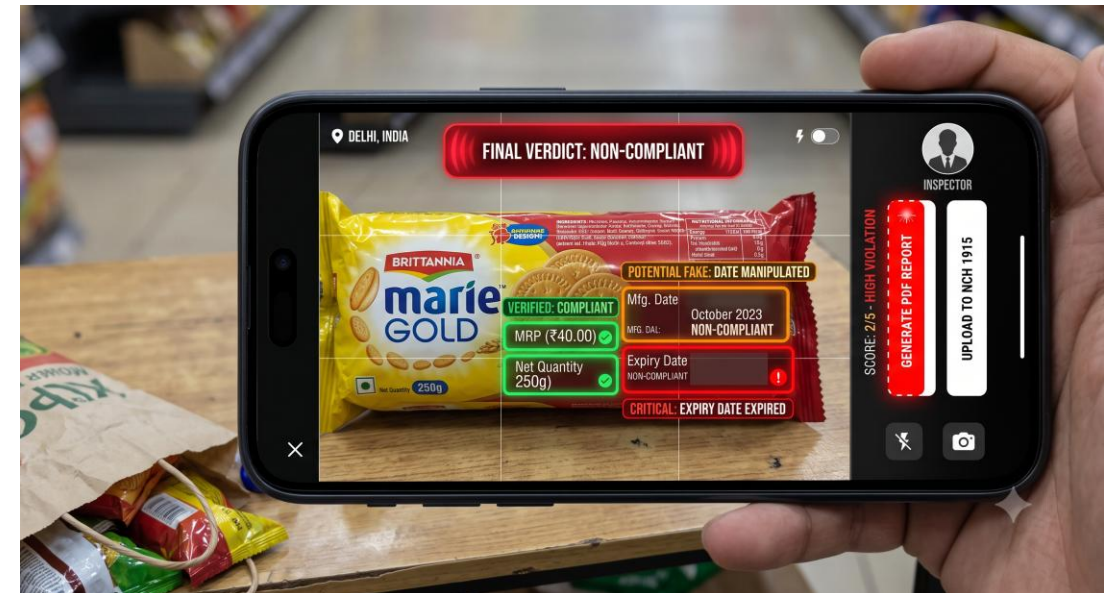
TITLE PAGE

- **Problem Statement ID – SIH26034**
- **Problem Statement Title-** Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels.
- **Theme-** Miscellaneous
- **PS Category-** Software
- **Team ID-** [Your Team ID]
- **Team Name (Registered on portal)-** The Hippos

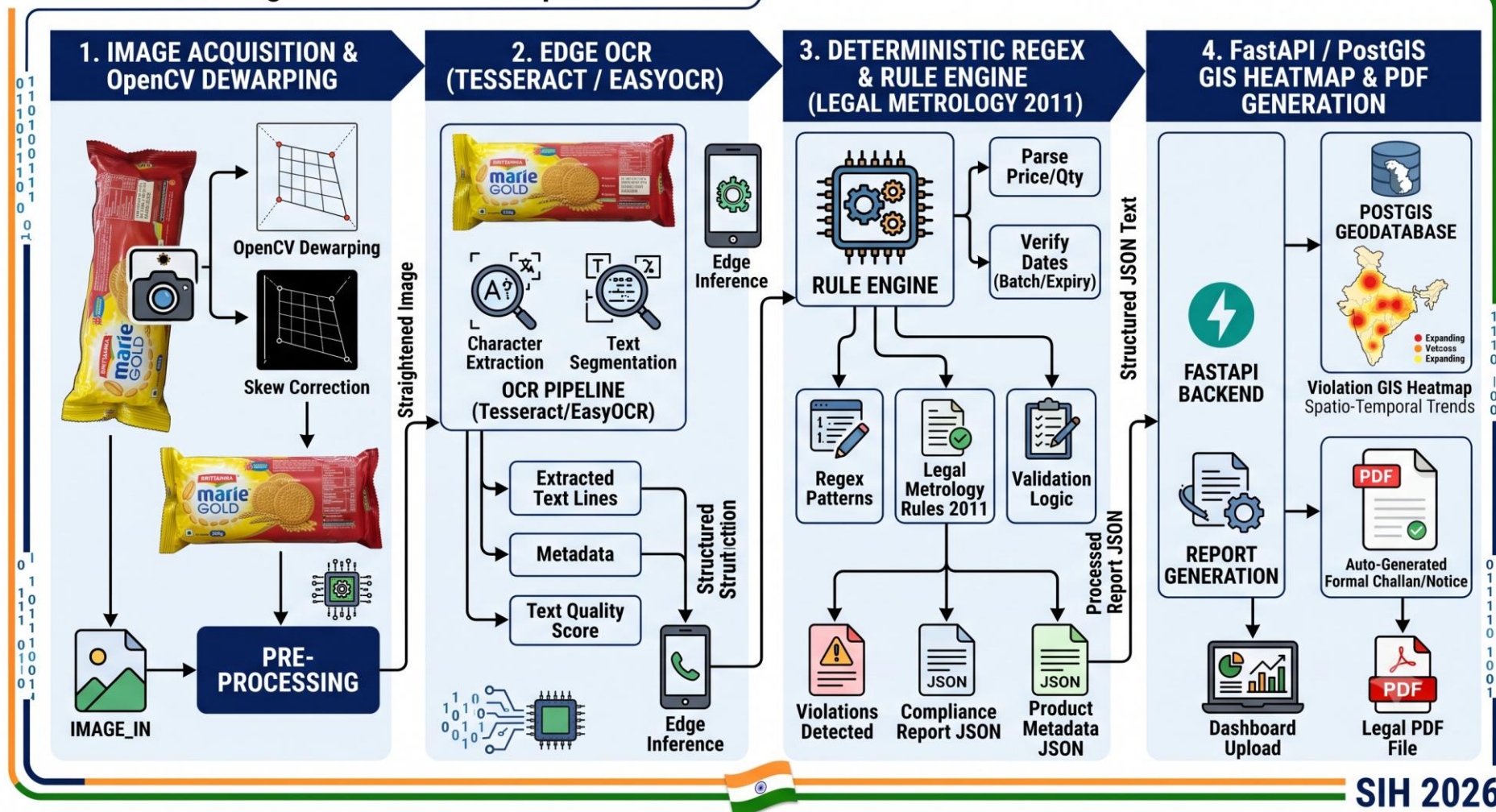


❖ Proposed Solution: Packaged Commodities Compliance Scanner

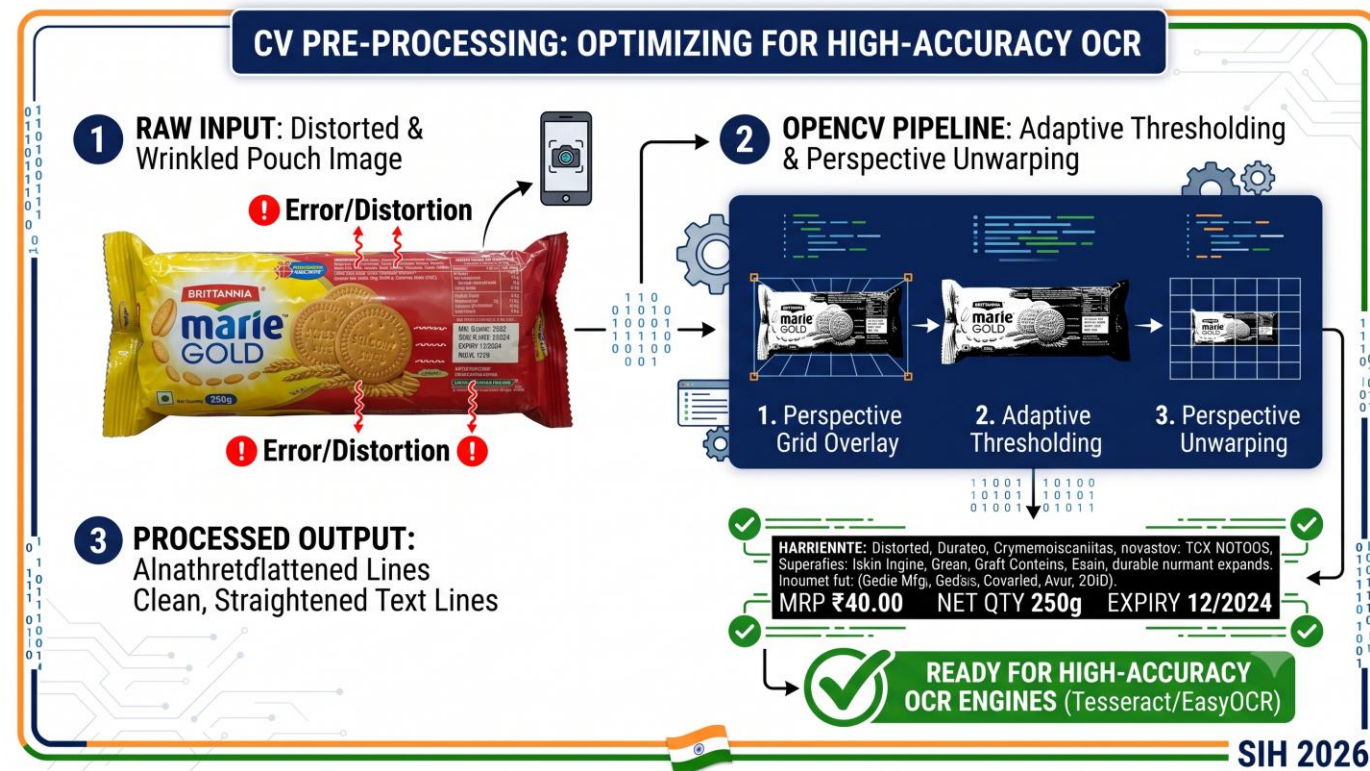
- A mobile app where an inspector or consumer points the camera at a product label; OCR extracts MRP, mfg. date, net quantity & consumer-care info, a rule engine checks it against the Legal Metrology Rules, 2011, and returns "COMPLIANT" or "MISSING: MRP not declared" with a court-ready PDF report and photo proof
- Replaces slow, manual, one-product-at-a-time inspection with an instant, camera-based compliance check that any inspector or citizen can run in seconds, across any shop
- Works fully offline (on-device OCR + local rule engine), auto-generates forensic evidence (photo + PDF report), and includes a batch "Inspector Mode" for scanning multiple products during a single raid



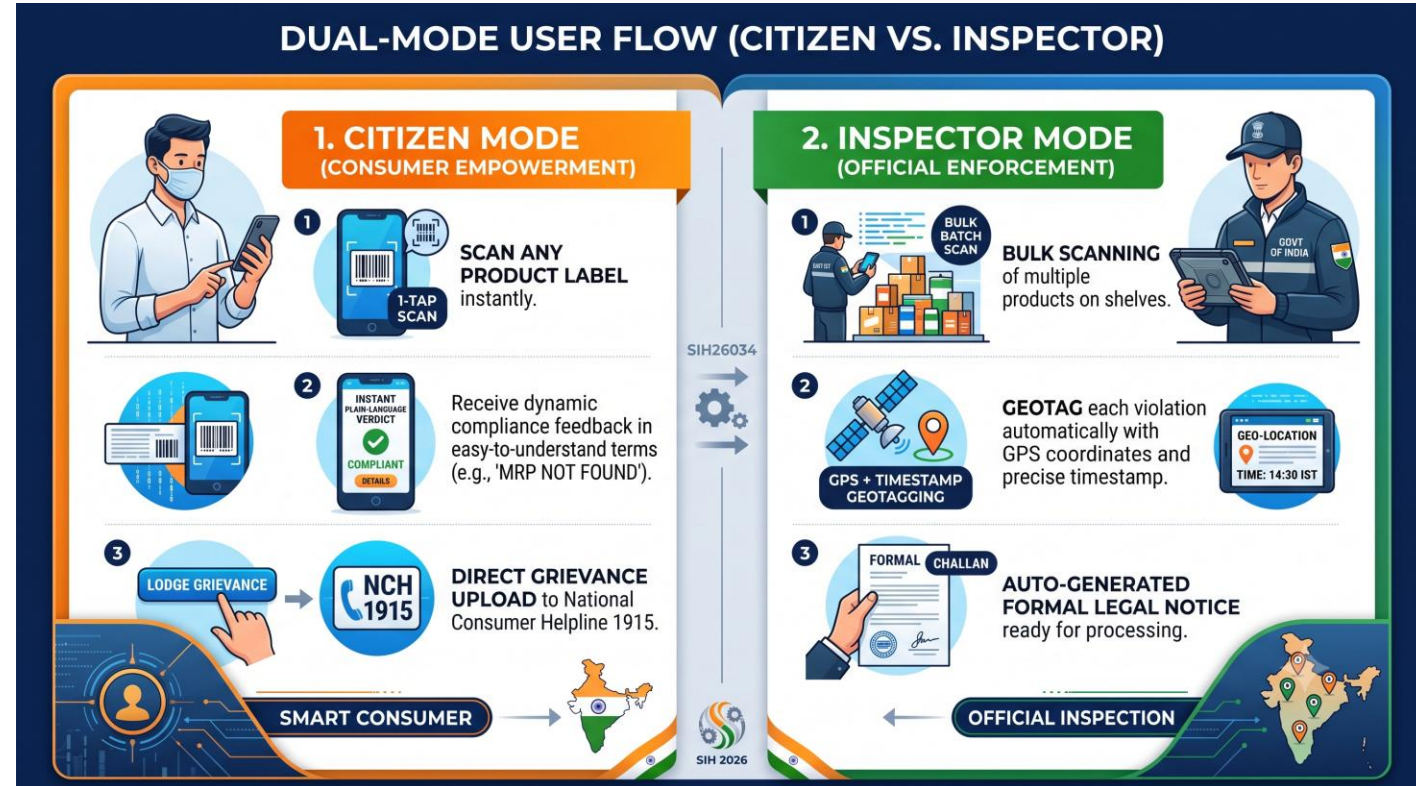
SIH26034 | Packaged Commodities Compliance Scanner



- Reuses an existing OCR + Celery pipeline built for our teacher-evaluation platform, so the core scanning flow ports in hours, not days; team already knows FastAPI/PostgreSQL/Flutter; test data (products) is free and available at any store
- OCR accuracy drops on curved, wrinkled or glare-heavy labels; label layouts vary across manufacturers and regional languages
- Image pre-processing (deskew, contrast correction) before OCR, manual-correction fallback for low-confidence fields, and an iteratively expanded rule library to handle new label formats



- Faster, wider Legal Metrology enforcement for Consumer Affairs inspectors, and empowers everyday consumers to self-check products before purchase, reducing mislabeled, expired or underweight goods reaching them
- Social: stronger consumer trust and protection. Economic: lowers manual inspection cost and speeds audits. Public health: cuts circulation of expired/mislabeled food & medicine. Works offline, so it scales to rural, low-connectivity markets



- Legal Metrology Act, 2009 & Legal Metrology (Packaged Commodities) Rules, 2011 — Dept. of Consumer Affairs, Govt. of India (legalmetrology.gov.in)
- Tesseract OCR / EasyOCR documentation — open-source OCR engines used for label text extraction
- Team's existing OCR + Celery pipeline from the teacher-evaluation platform — internal reference implementation being reused

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.